A measured floor on index-based organ dose estimation in CT, where per-organ coefficients do not improve on a single conversion factor

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Abstract

Objective. The volume CT dose index is a scanner output, not a patient dose, and the field converts it with size corrections, dose-length-product factors and per-organ coefficient libraries. How close such a conversion can get has not been measured against transport on the same anatomy. We measure that floor.

Approach. A GPU Monte-Carlo pipeline computes absorbed organ dose in patient anatomy from routine metadata alone. The transport engine is verified against AAPM TG-195 Case 4 with a replica-measured uncertainty and a yardstick fixed beforehand. On 30 multi-vendor abdominal CT series from a public archive, 342 organ-series pairs across 12 organs, we ask how well absorbed dose is recovered from an organ-weighted CTDIvol index by conversions of increasing richness, each scored under leave-one-patient-out cross-validation.

Main results. At the centre volume of interest the engine differs from PENELOPE by 0.359 %, or 1.67 combined standard deviations. A single conversion factor for the cohort recovers absorbed organ dose to a median 31 %. One factor per organ does not improve on it (32 %). Adding a scan-length term reaches 16 %, and adding either organ identity or patient size reaches a floor of 13 % and 13 %. 88% of the variance in the index-to-dose ratio lies between patients; 6% lies between organs. Monte-Carlo statistical uncertainty is 0.012 % of the dose at the median, so the residual is physical.

Significance. The information an index withholds is a property of the patient, not of the organ. A per-organ coefficient table cannot supply it, and neither can a size correction; on this cohort the best index-based conversion still misses absorbed organ dose by a median of 13 % and by 32 % at the ninetieth percentile.

Keywords: computed tomography, organ dose, Monte Carlo, CTDIvol, dose index, cross-validation

1. Introduction

The volume CT dose index is a property of the scanner and a standard phantom, not of the patient in it, and saying so has been necessary often enough to have its own paper (McCollough et al 2011). It is nonetheless the number every scanner reports, the number archived in dose-monitoring systems, and therefore the number from which patient dose is estimated in practice (McCollough et al 2008).

The field has three established ways of getting from that index to something closer to a dose. Size-specific dose estimates rescale CTDIvol by the patient's attenuating size (Boone et al 2011), with water-equivalent diameter as the size that matters (McCollough et al 2014). Dose-length product is converted to effective dose by protocol-specific factors (Deak et al 2010). And organ dose is estimated from libraries of per-organ coefficients, computed once by Monte Carlo on reference phantoms and looked up thereafter (Lee et al 2011, Lee et al 2015), including under tube current modulation (Tian et al 2015).

All three are conversions from an index. Each replaces transport through this patient's anatomy with a coefficient obtained from someone else's. The question this paper asks is not whether such a conversion is exact — it is not, and no one claims it is — but how close it can get, and what the irreducible part is made of.

That question needs absorbed organ dose on the same anatomy as the index, which is what we compute. A companion study reports an organ-weighted CTDIvol index: CTDIvol multiplied by the tube-current mean over each segmented organ relative to the scan mean (Yamamoto 2026a). It is an index, and a good one: it is organ-specific and it carries the longitudinal modulation. It does not transport photons. Against it we place absorbed dose in milligray, computed by Monte Carlo through each patient's own segmented, density-mapped anatomy, with every dose coefficient computed rather than looked up.

We then fit conversions of increasing richness from the index to the dose — a single factor, a per-organ factor, a scan-length term, and combinations — and score each on patients the fit has not seen. The result is a floor, and a decomposition of what sits beneath it.

What this paper does not claim. It does not claim that an index and an absorbed dose are the same quantity, nor that anyone believes they are. It does not claim novelty for GPU Monte-Carlo photon transport (Badal and Badano 2009), for

spectrum modelling (Bujila et al 2020, Poludniowski et al 2021), for HU-to-density calibration (Schneider U et al 1996, Schneider W et al 2000), or for deep-learning organ segmentation (Wasserthal et al 2023); each is used as published. It does not claim that a scan-length term is a discovery: CTDIvol is defined per rotation and carries no length, dose-length product is the standard quantity that does, and the fitted length dependence is close to proportional. The term is included because a conversion that omitted it would be a straw man, and its slope is reported so that a reader can judge that for themselves.

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2. Methods

2.1 Monte-Carlo engine and build provenance

Photon transport uses MC-GPU (Badal and Badano 2009), a public-domain voxelised Monte-Carlo code, built locally. The build is pinned: the source patches applied, their kind, and the SHA-256 of the resulting binary are recorded in a build manifest that ships with the software. Patches are of three kinds — defect fixes, portability changes, and capacity increases that raise a compile-time array bound without touching the physics — and each capacity patch records the bound, the new value, and why it was needed. Two were required here: the Schneider tissue groups at 140 kVp exceed the upstream limits on Compton shells and on Rayleigh energy bins.

Material cross-sections are generated locally from the photon subset of the PENELOPE 2006 database (Salvat et al 2006) rather than transcribed, so that no coefficient table is redistributed and each material file carries the digest of the data it came from.

2.2 Source model

X-ray spectra come from SpekPy (Bujila et al 2020, Poludniowski et al 2021), which supplies the spectrum, the half-value layer and the absolute tube output for each tube potential. The source is modelled as an equivalent spectrum and filtration in the manner of Turner *et al* (2009), with a bowtie profile represented as a set of sub-fans, each carrying its own spectrum. The profiles are derived from the published measurements of Yang *et al* (2019), which cover three vendors; the fourth is represented by the cross-vendor average with its spread declared. Geometry, collimation, source-to-axis distance and the number of projections come from the series metadata.

2.3 Verification against TG-195 Case 4

The engine is verified against Case 4 of AAPM Task Group 195 (Sechopoulos et al 2015), the voxelised phantom case with published results from four independent codes. Two properties of this verification matter for what follows.

The yardstick was fixed before any comparison was run. PENELOPE was chosen because it shares the photon cross-section database from which our material files are generated, not because it agreed best; the other three codes are reported alongside so that the choice is visible.

The uncertainty is measured rather than assumed. Ten independent replicas of the anchor configuration give a sample standard deviation, from which the standard error of the anchor follows, and this is combined with the reported reference uncertainty to give a combined standard deviation against which the difference is expressed. Agreement is reported with that uncertainty; it is not converted into a pass or a fail against a threshold of our own choosing.

The phantom length term was measured, not bounded: a 40 cm phantom is indistinguishable from a 160 cm phantom at these volumes of interest.

2.4 Cohort selection

Series were selected from a public archive (Clark et al 2013, Moawad et al 2023) by a recorded funnel that begins at the collection index and ends at the series carried forward, with every exclusion counted and named (figure 1). The selection is re-derivable: the criteria are in code, and the surviving Series Instance UIDs, their collection and their licence are recorded so that any series can be re-fetched. No imaging is redistributed.

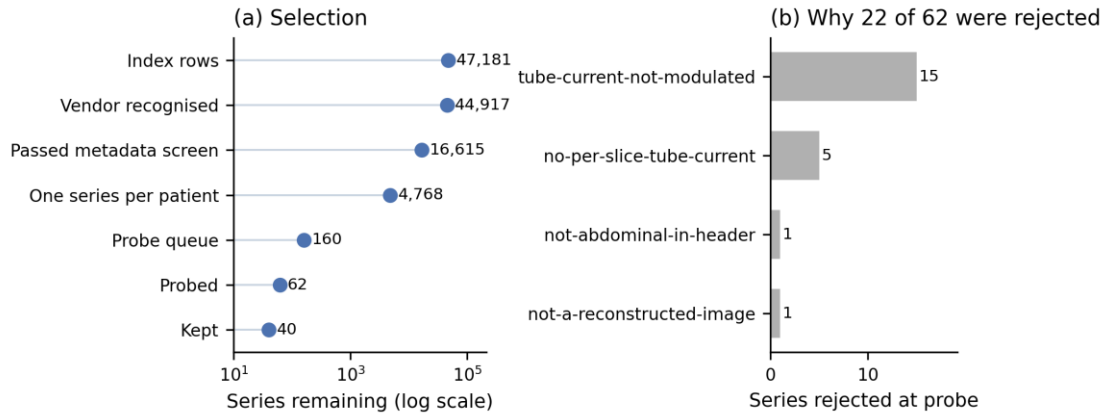


Figure 1. Cohort selection. (a) Series remaining at each stage, from the 47 181 rows of the collection index to the 40 series carried forward, on a logarithmic axis. Position

carries the count: a bar length on a logarithmic axis would encode the ratio to the axis floor rather than the number. (b) The reason each of the 22 probed-and-rejected series was rejected; almost all lacked the per-slice tube current the pipeline requires

A metadata screen removes series that are not diagnostic reconstructed abdominal images or that carry too few or too many slices; one series per patient is kept; the remainder are probed in vendor-balanced order until ten series per vendor are accepted. The probe reads the header and rejects a series that lacks the per-slice tube current the pipeline requires, which accounts for most of the rejections (figure 1b).

2.5 Eligibility: a pre-declared density gate

Voxels inside the body contour at or above the top of the density calibration are metal, concentrated contrast, or artefact, and are not tissue. A series is excluded if its largest connected out-of-calibration component exceeds 1 cm^3. The threshold and the criterion were declared before the volumes were computed. Of 40 assessed series, 32 are eligible and 8 are excluded (figure 2). The distance from the offending component to the nearest scored organ is reported but is not part of the gate.

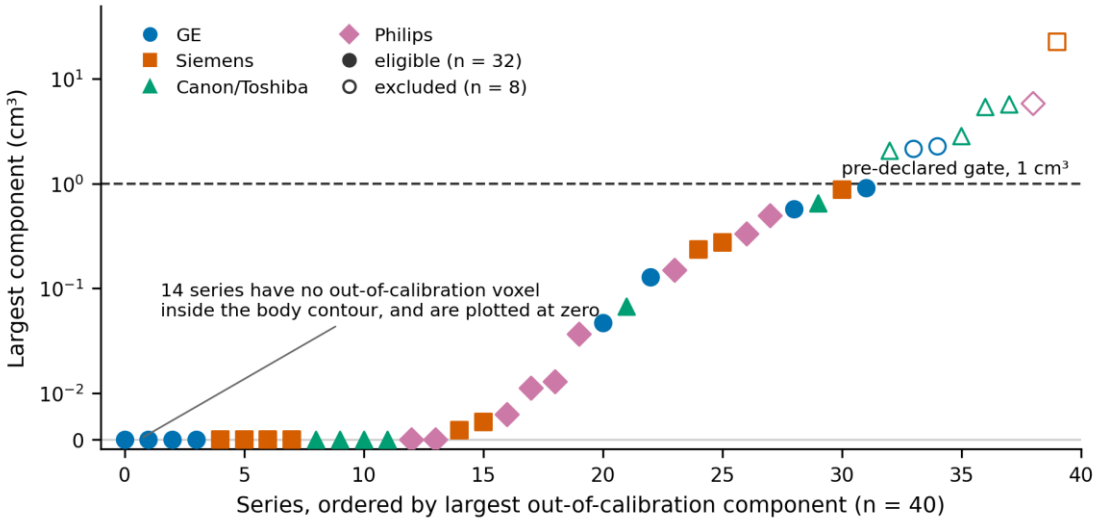


Figure 2. The pre-declared density gate. Largest out-of-calibration component per series — voxels inside the body contour at or above the calibration ceiling — for all 40 assessed series, ordered by that volume, on a symmetric-logarithmic axis so that the 14 series with no such voxel are visible rather than dropped. Filled markers are the 32 eligible series, open markers the 8 excluded. The gate at 1 cm^3 was declared before the volumes were computed

2.6 Segmentation, density and materials

Organ boundaries come from TotalSegmentator (Wasserthal et al 2023) run at inference on each series, so that no manual contour enters the pipeline and the same procedure applies to every vendor. Hounsfield units are mapped to mass density and to Schneider tissue groups (Schneider U et al 1996, Schneider W et al 2000); voxels above the top calibration breakpoint are held at its density and counted, and it is that count the eligibility gate reads.

2.7 CTDIvol normalisation

Absorbed dose per emitted photon is converted to dose per examination by a scale derived from a simulated CTDI phantom measurement at the same tube potential and vendor, matched to the CTDIvol the series carries. Where the series records a CTDIvol, that value is used; where it does not, the value reconstructed from scanner model and technique by a companion tool is used, and the two are distinguished throughout (Yamamoto 2026b). Of 32 eligible series, 23 carry a recorded CTDIvol, 7 carry a reconstructed one, and 2 carry neither and are omitted from every analysis that needs one.

2.8 Organ dose

Dose is tallied on an isotropic grid over the scan-covering geometry, with the number of longitudinal stations set by the scan length and the nominal collimation, and is reported per organ as the mass-weighted mean over the segmented voxels of that organ, with its Monte-Carlo statistical uncertainty.

2.9 Index-based conversions, and how they are scored

Let r be the ratio of absorbed organ dose to the organ-weighted CTDIvol index for one organ in one series. Five conversions are compared, each predicting r and therefore predicting the dose from the index:

1. **a single factor** for the whole cohort;
2. **one factor per organ**, which is what a per-organ coefficient table offers;
3. **a scan-length term**, fitted as $\log r = a + b \log n_z$ on the number of longitudinal stations;
4. **scan length and a per-organ offset**, which is everything an index can carry;
5. **scan length and patient size**, with size as the effective diameter derived from the segmented body volume and the scan length.

All are fitted in log ratio, because the quantity is a multiplicative factor spanning several fold and its errors are relative.

Each is scored under leave-one-patient-out cross-validation: for each series, the conversion is fitted on the other 30 minus one series and evaluated on the held-out one. Fitting twelve organ offsets on thirty patients and reporting the error on those same thirty is optimistic by construction, and thirty is small enough for that to matter. Every error reported below is on a patient the fit did not see.

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3. Results

3.1 Verification

At the centre volume of interest the engine differs from PENELOPE by 0.359 %, which is 1.67 combined standard deviations (figure 3). The difference from the other three published codes is larger and in the same direction at the centre, and smaller at the perimeter; showing all four is what makes the choice of yardstick auditable rather than convenient. The phantom length term is measured and null.

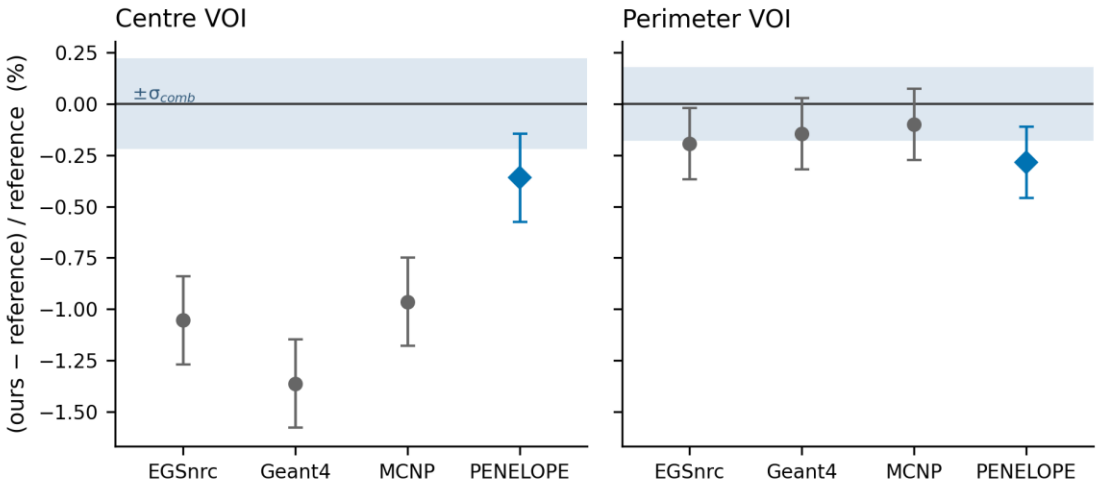


Figure 3. Verification against AAPM TG-195 Case 4, at the centre and perimeter volumes of interest. Each point is the relative difference between this engine and one published code, with error bars of the combined standard uncertainty. PENELOPE, marked with a diamond, is the yardstick, and was chosen before any comparison was run because it shares the photon cross-section database this work generates its materials from; the other three codes are shown so the choice can be seen not to have been made after the fact

3.2 Cohort

47181 index rows reduce to 40 series through the funnel of figure 1, of which 62 were probed and 22 rejected. The density gate leaves 32 eligible series (figure 2), and 30 of those carry a CTDIvol and enter the analysis.

3.3 Absorbed organ doses

Across 342 organ-series pairs the median absorbed organ dose is 24.3 mGy, with a range of 3.9 to 84.0 mGy. Per-organ values by vendor are given in table 1 and figure 6; with six to nine series per vendor the cohort is not powered for a vendor comparison and none is offered.

Figure 5 Scan-covering organ dose by vendor

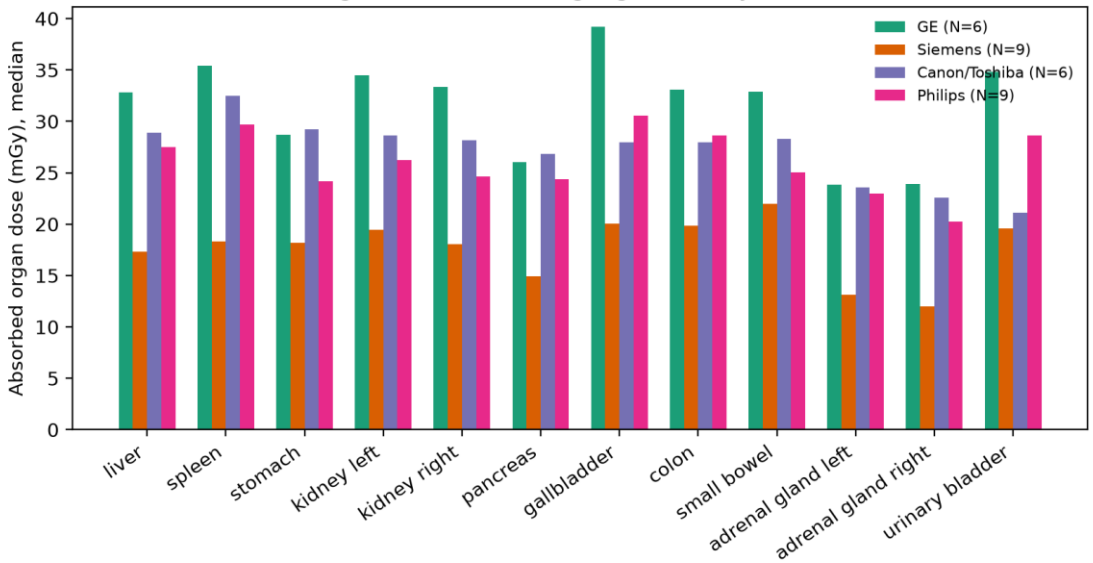


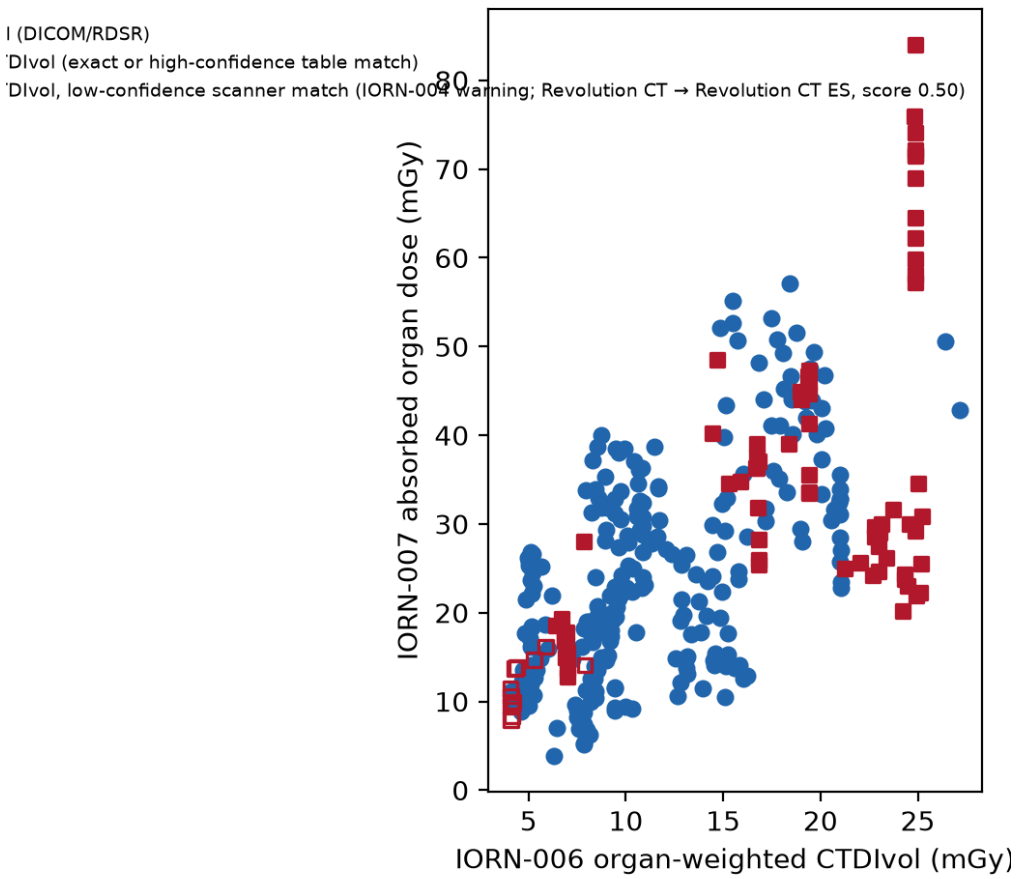
Figure 6. Absorbed organ dose by vendor, per organ, with the number of series in which each organ is present. Descriptive: with six to nine series per vendor the cohort is not powered for a vendor comparison, and none is claimed

Organ	GE	Siemens	Canon/Toshiba	Philips	All series
Liver	32.8 (6)	17.3 (8)	28.9 (6)	27.5 (9)	27.5 [16.2, 33.9] (29)
Spleen	35.4 (6)	18.3 (8)	32.5 (6)	29.7 (9)	29.7 [16.8, 37.1] (29)
Stomach	28.7 (6)	18.2 (8)	29.3 (6)	24.2 (9)	24.2 [16.7, 32.9] (29)
Pancreas	26.0 (6)	14.9 (8)	26.8 (6)	24.4 (9)	23.8 [14.4, 30.3] (29)
Gallbladder	39.2 (4)	20.0 (7)	28.0 (6)	30.6 (8)	28.8 [14.9, 34.0] (25)
Kidney, left	34.5 (6)	19.4 (8)	28.6 (6)	26.2 (9)	26.2 [17.8, 38.7] (29)
Kidney, right	33.3 (6)	18.0 (9)	28.2 (6)	24.6 (9)	25.2 [15.6, 37.5] (30)
Adrenal, left	23.8 (6)	13.1 (8)	23.6 (6)	23.0 (9)	22.3 [12.6, 28.0] (29)
Adrenal, right	23.9 (6)	12.0 (8)	22.6 (6)	20.2 (9)	21.5 [11.5, 25.9] (29)
Small bowel	32.9 (6)	22.0 (9)	28.3 (6)	25.0 (9)	24.6 [16.7, 35.6] (30)
Colon	33.1 (6)	19.8 (9)	27.9 (6)	28.6 (9)	24.8 [18.8, 35.1] (30)
Bladder	34.8 (5)	19.6 (8)	21.1 (6)	28.6 (5)	25.3 [17.8, 29.1] (24)

Table 1. Absorbed organ dose in mGy over the 30 series carrying a CTDIvol. Vendor columns give the median and, in parentheses, the number of series in which that organ is present; the final column pools every series and adds the interquartile range. Series per vendor: GE 6, Siemens 9, Canon/Toshiba 6, Philips 9. The cohort is not powered for a vendor comparison and none is claimed.

Plotted against the organ-weighted index on the same anatomy (figure 4), the two are correlated — Pearson r = 0.66 over all 30 series and 0.61 over the 23 with a recorded

CTDIvol — and the median ratio of dose to index is 2.17. That the two differ is not the result; how far apart they remain after every available correction is.



Open squares: CTDIvol reconstructed under IORN-004's own low-confidence warning (Revolution CT matched to Revolution CT ES, score 0.50). Filled squares: reconstructed with an exact or high-confidence table match. Filled circles: recorded in DICOM/RDSR. Two series omitted (LightSpeed Ultra, Revolution EVO).

Figure 4. Same-patient scatter: the organ-weighted CTDIvol index against absorbed organ dose in mGy, one point per organ per series. 12 organs, 342 points, 30 series. Filled circles are the 23 series carrying a recorded CTDIvol, filled squares the 6 reconstructed on an exact or high-confidence table match, open squares the 1 reconstructed under a low-confidence warning. The two quantities are correlated and are not the same quantity; how far apart they remain after every available correction is Figure 5

3.4 The floor on an index-based conversion

Figure 5a gives the held-out error of each conversion. A single factor for the whole cohort recovers absorbed organ dose to a median 31 %, with 75 % at the ninetieth percentile.

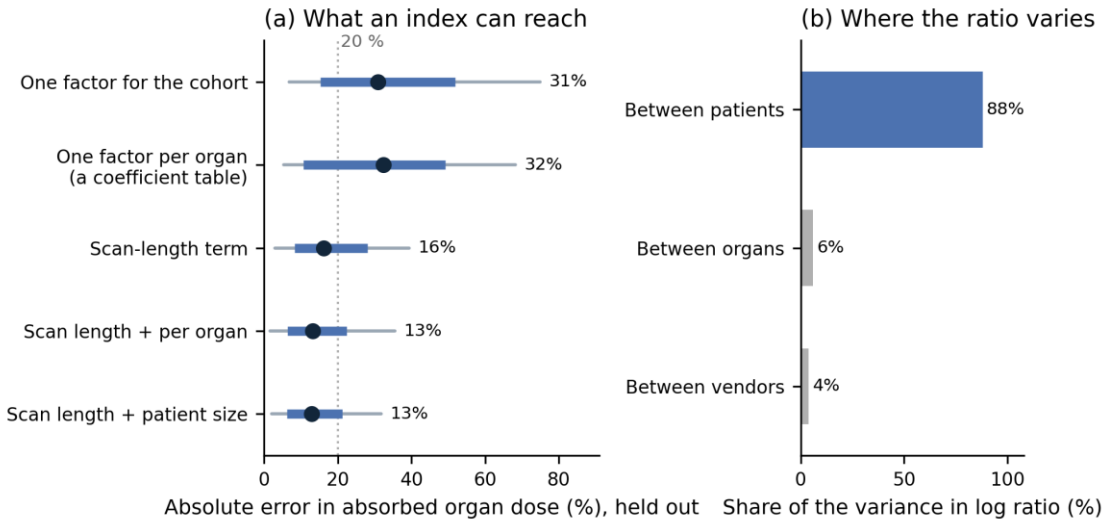


Figure 5. The primary endpoint. (a) Absolute error in absorbed organ dose for each index-based conversion, under leave-one-patient-out cross-validation: point is the median, thick bar the interquartile range, thin bar the 10th to 90th percentile. A per-organ coefficient does not improve on a single factor for the whole cohort. A scan-length term does, and adding either the organ or the patient's size to it changes little: the floor is reached at a median of 13 %. (b) Where the index-to-dose ratio varies. Almost all of it is between patients; organ identity and vendor together account for under a tenth

One factor per organ does not improve on it. Cross-validated, the per-organ conversion is slightly worse — a median 32 % against 31 % — and the fraction of organs recovered to within 20 % moves from 35% to 37%. Twelve organ means fitted on twenty-nine patients do not transfer to the thirtieth. This is the form in which published organ-dose coefficient libraries are used (Lee et al 2011, Lee et al 2015), and on this cohort that form carries no information a single number does not.

A scan-length term does improve on it, to a median 16 % and 39 % at the ninetieth percentile. This is expected rather than surprising: CTDIvol is defined per rotation, dose-length product is the quantity that carries length (Deak et al 2010), and the fitted slope on $\log n_z$ is 1.14 — close enough to one that the dependence is largely accumulation.

Adding to the length term everything else an index could know changes little. A per-organ offset on top of length reaches 13 %, and the patient's effective diameter reaches 13 %, with 32 % at the ninetieth percentile and 72% of organs inside 20 %. That is the floor: nothing an index carries closes the remaining gap.

3.5 The residual is a property of the patient

Figure 5b decomposes the variance of the log index-to-dose ratio. 88% of it lies between patients, 6% between organs and 4% between vendors.

The same fact stated the other way: within one patient, a single number converts the index to absorbed dose for every one of that patient's organs to within a factor of 1.18; across patients that number ranges from 0.88 to 4.79, a span of 5.5-fold. The information an index withholds is not organ-specific. It is patient-specific, and it is nearly common to all of that patient's organs at once.

This explains section 3.4 completely. A per-organ coefficient addresses the 6%; it cannot address the 88%, and cross-validated it does not even secure the 6%.

3.6 Patient size does not stand in for it

The obvious candidate for a patient-level correction is patient size, which is what size-specific dose estimates apply (Boone et al 2011, McCollough et al 2014). It does not carry this. The correlation between the patient-level factor and the effective diameter is Pearson r = -0.30 in log, over a cohort spanning 23.5 to 38.7 cm, and adding size on top of the length term moves the median error from 16 % to 13 %, the same place a per-organ offset reaches.

3.7 The residual is physical, not statistical

Per-organ Monte-Carlo statistical uncertainty is 0.012 % of the organ dose at the median and 0.83 % at worst. The floor of section 3.4 is more than three orders of magnitude above it. The scatter that no index-based conversion removes is a property of the transport, not of the sampling.

3.8 Robustness

Restricting the analysis to the 23 series that carry a recorded CTDIvol, rather than a reconstructed one, does not change the index-to-dose relationship: the correlation is 0.61 against 0.66, and the median ratio is 2.17 against 2.17. No additional Monte Carlo was run for this check.

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4. Discussion

The result is a limit on a class of methods, and it is quantitative. Every conversion from a dose index to an organ dose — whether by a size correction, a length factor or a coefficient table — is estimating a patient-level quantity from measurements that do not contain it. On this cohort, the best such conversion misses absorbed organ

dose by a median of 13 % and by 32 % at the ninetieth percentile, and the part it misses is nearly the same multiplicative error for every organ in a given patient.

That last property is the useful one, and it cuts both ways. It means an index is a sound instrument for questions internal to a patient: which organ received more, how one protocol compares with another on the same person, whether a change in modulation moved dose from one organ to another. Within a patient the conversion factor is constant to within 1.18-fold, so those comparisons survive. It also means an index is a poor instrument for questions across patients in absolute milligray, which is exactly what epidemiological and risk-estimation uses require, because there the 5.5-fold between-patient span is the whole of the signal being estimated.

The negative result on per-organ coefficients deserves stating carefully. It does not say that published coefficient libraries (Lee et al 2011, Lee et al 2015) are wrong; they are computed on reference phantoms and are correct for those phantoms. It says that the residual they are asked to absorb, when applied to a real patient through an index, is not organ structured. Cross-validated on this cohort, twelve fitted organ coefficients did not beat one number — and a table with more coefficients, fitted on more organs, would be fitting more finely into the 6% of variance that organ identity explains.

Whether the floor found here is the floor everywhere is not something one cohort can settle. It is a floor on abdominal CT, on twelve abdominal organs, in a cohort selected for having per-slice tube current recorded. What is more likely to generalise is the structure of the residual rather than its size: an index conditions on the scanner and the protocol, and the transport it omits is a function of the patient.

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5. Limitations

The cohort is 30 series across four vendors, six to nine per vendor. That is adequate for the patient-level question, which has 30 independent units, and inadequate for a vendor-level one, which would have four. No vendor claim is made; table 1 and figure 6 are descriptive.

There is no ground-truth absorbed dose in a patient. The engine is verified against a published reference phantom case (Sechopoulos et al 2015) with a measured uncertainty, and the patient doses inherit that verification, but they are not independently measured. What the verification supports is the transport; what it cannot support is the segmentation, the HU-to-density mapping, or the source model as applied to a particular scanner.

Organ boundaries come from an automatic segmentation (Wasserthal et al 2023) applied uniformly. Its errors propagate into both the index and the dose, since both are computed over the same masks, and would tend to reduce rather than inflate the difference between them.

The eligibility gate excludes series with more than 1 cm^3 of out-of-calibration material. Patients with implants are therefore under-represented, and the floor reported here is a floor for anatomy the density calibration covers.

The scan-length term is fitted on the number of longitudinal stations, which is derived from the scan length and the nominal collimation, not on a directly recorded irradiated length. The fitted slope of 1.14 is reported so that the reader can see how much of that term is accumulation.

Doses are computed for the acquisition as reconstructed and recorded. Over-ranging, localiser exposures and any repeat series are outside the scope of the metadata used.

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6. Conclusions

Absorbed organ dose in CT can be recovered from an organ-weighted CTDIvol index to a median of 13 % and no better, held out on patients the conversion has not seen. A per-organ coefficient does not improve on a single conversion factor for the whole cohort, because 88% of the variance in the index-to-dose ratio is between patients and only 6% is between organs. Patient size, the standard patient-level correction, does not carry it either. The residual is more than three orders of magnitude above the Monte-Carlo statistical uncertainty, so it is physical: it is the transport that an index does not perform.

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Data availability

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.5281/zenodo.22233153.

That archive is the release of the software described here, under the MIT licence, and holds the frozen result tables and the commands that regenerate every figure; development continues at `https://github.com/Institute-of-One/ct-dosemc-core`. No imaging is redistributed: the collection, Series Instance UID, licence and retrieval date of every series are recorded so that each can be re-fetched from the archive (Clark et al 2013, Moawad et al 2023). Every number reported here is resolved at build time from a machine-readable result file, and the build fails if a number cannot be traced to one.

References

Badal A and Badano A 2009 Accelerating Monte Carlo simulations of photon transport in a voxelized geometry using a massively parallel Graphics Processing Unit *Med. Phys.* **36** 4878–4880 [doi:10.1118/1.3231824]

Boone J M, Strauss K J, Cody D D, McCollough C H, McNitt-Gray M F and Toth T L 2011 *Size-Specific Dose Estimates (SSDE) in Pediatric and Adult Body CT Examinations* AAPM Report 204 (College Park, MD: American Association of Physicists in Medicine) [doi:10.37206/143]

Bujila R, Omar A and Poludniowski G 2020 A validation of SpekPy: a software toolkit for modelling X-ray tube spectra *Phys. Med.* **75** 44–54 [doi:10.1016/j.ejmp.2020.04.026]

Clark K, Vendt B, Smith K, Freymann J, Kirby J, Koppel P, Moore S, Phillips S, Maffitt D, Pringle M, Tarbox L and Prior F 2013 The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository *J. Digit. Imaging* **26** 1045–1057 [doi:10.1007/s10278-013-9622-7]

Deak P D, Smal Y and Kalender W A 2010 Multisection CT protocols: sex- and age-specific conversion factors used to determine effective dose from dose-length product *Radiology* **257** 158–166 [doi:10.1148/radiol.10100047]

Lee C, Kim K P, Long D, Fisher R, Tien C, Simon S L, Bouville A and Bolch W E 2011 Organ doses for reference adult male and female undergoing computed tomography estimated by Monte Carlo simulations *Med. Phys.* **38** 1196–1206 [doi:10.1118/1.3544658]

Lee C, Kim K P, Bolch W E, Moroz B E and Folio L 2015 NCICT: a computational solution to estimate organ doses for pediatric and adult patients undergoing CT scans *J. Radiol. Prot.* **35** 891–909 [doi:10.1088/0952-4746/35/4/891]

McCollough C H, Cody D D, Edyvean S, Geise R, Gould B, Keat N, Huda W, Judy P, Kalender W, McNitt-Gray M, Morin R, Payne T, Stern S, Rothenberg L, Shrimpton P, Timmer J and Wilson C 2008 *The Measurement, Reporting, and Management of Radiation Dose in CT* AAPM Report 96 (College Park, MD: American Association of Physicists in Medicine) [doi:10.37206/97]

McCollough C H, Leng S, Yu L, Cody D D, Boone J M and McNitt-Gray M F 2011 CT dose index and patient dose: they are not the same thing *Radiology* **259** 311–316 [doi:10.1148/radiol.11101800]

McCollough C H, Bakalyar D M, Bostani M, Brady S, Boedeker K, Boone J M, Chen-Mayer H H, Christianson O I, Leng S, Li B, McNitt-Gray M F, Nilsen R A, Supanich M P and Wang J 2014 *Use of Water Equivalent Diameter for Calculating Patient Size and Size-Specific Dose Estimates (SSDE) in CT* AAPM Report 220 (College Park, MD: American Association of Physicists in Medicine) [doi:10.37206/146]

Moawad A M, Fuentes D, ElBanan M M, Shalaby A S, Guccione J, Kamel S, Jensen C T and Elsayes K M 2023 Multimodality annotated hepatocellular carcinoma data set including pre- and post-TACE with imaging segmentation *Sci. Data* **10** 33 [doi:10.1038/s41597-023-01928-3]

Poludniowski G, Omar A, Bujila R and Andreo P 2021 Technical Note: SpekPy v2.0 — a software toolkit for modeling x-ray tube spectra *Med. Phys.* **48** 3630–3637 [doi:10.1002/mp.14945]

Salvat F, Fernández-Varea J M and Sempau J 2006 *PENELOPE — a code system for Monte Carlo simulation of electron and photon transport* (Issy-les-Moulineaux: NEA-OECD)

Schneider U, Pedroni E and Lomax A 1996 The calibration of CT Hounsfield units for radiotherapy treatment planning *Phys. Med. Biol.* **41** 111–124 [doi:10.1088/0031-9155/41/1/009]

Schneider W, Bortfeld T and Schlegel W 2000 Correlation between CT numbers and tissue parameters needed for Monte Carlo simulations of clinical dose distributions *Phys. Med. Biol.* **45** 459–478 [doi:10.1088/0031-9155/45/2/314]

Sechopoulos I, Ali E S M, Badal A, Badano A, Boone J M, Kyprianou I S, Mainegra-Hing E, McMillan K L, McNitt-Gray M F, Rogers D W O, Samei E and Turner A C 2015 Monte Carlo reference data sets for imaging research: executive summary of the report of

AAPM Research Committee Task Group 195 *Med. Phys.* **42** 5679–5691 [doi:10.1118/1.4928676]

Tian X, Li X, Segars W P, Frush D P and Samei E 2015 Prospective estimation of organ dose in CT under tube current modulation *Med. Phys.* **42** 1575–1585 [doi:10.1118/1.4907955]

Turner A C, Zhang D, Kim H J, DeMarco J J, Cagnon C H, Angel E, Cody D D, Stevens D M, Primak A N, McCollough C H and McNitt-Gray M F 2009 A method to generate equivalent energy spectra and filtration models based on measurement for multidetector CT Monte Carlo dosimetry simulations *Med. Phys.* **36** 2154–2164 [doi:10.1118/1.3117683]

Wasserthal J, Breit H-C, Meyer M T, Pradella M, Hinck D, Sauter A W, Heye T, Boll D T, Cyriac J, Yang S, Bach M and Segeroth M 2023 TotalSegmentator: robust segmentation of 104 anatomic structures in CT images *Radiol. Artif. Intell.* **5** e230024 [doi:10.1148/ryai.230024]

Yamamoto S 2026a *ctsegdose-core: anatomy-weighted CTDIvol from routine CT metadata, using deep-learning organ segmentation* v0.1.1 (Zenodo) [doi:10.5281/zenodo.21817307]

Yamamoto S 2026b *ctdose-core: open, auditable CT dose surveillance from DICOM, with a physics reconstruction when the dose attributes are missing* v0.1.1 (Zenodo) [doi:10.5281/zenodo.21636719]

Yang K, Ruan C, Li X and Liu B 2019 Data of CT bow tie filter profiles from three modern CT scanners *Data in Brief* **25** 104261 [doi:10.1016/j.dib.2019.104261].
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